

SKILLS

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|----------------------------|---|-----------------------------|
| • Python C C++ | • Sensor Fusion Lasers RF Atomic System | • Schematic + PCB Design |
| • RTOS ARM Cortex FPGA | • Mission Critical SW Real-Time Control | • System Simulation |
| • Robotics Simulation | • Edge Detection Low Latency (Vision) UAVs | • Directed Full-Stack Teams |
| • Control Systems | • Digital Signal Processing VHDL SoC | • DevOps CI/CD Pipelines |

WORK EXPERIENCE

Senior Software Engineer | C++, BirdDog, Melbourne, Australia *Aug 2023 – Mar 2025*

- Led scope, planning, design of video broadcasting (low latency) encoder and decoders with 2 teams.

Senior (Lead Electrical) Engineer | C, Frequency Electronics, New York, New York *Aug 2022- July 2023*

- Designed and implemented critical control systems for Rubidium Atomic Clock
 - Designed the core Timing, RF, laser, and clock control systems for the next generation Pulsed Atomic Rubidium Frequency Standard utilizing the Xilinx ZCU216 and KR260.
 - RF System: Synthesized a pulsed, FSK, and ASK 6.8 GHz RF signal to be steered towards the rubidium hyperfine transition via the Xilinx RF Data Converter and custom VHDL on a RF System on Chip (SoC).

Electrical Engineer | VHDL (FPGA) | C, Frequency Electronics, New York, New York *May 2020- Aug 2022*

- Formulated control systems for a multi-million-dollar military Office of Naval Research program.
 - Lead RF SoC development. (real-time processing, FPGA development, & application processing)

Electrical Engineer | Co-Founder | Autonomy, Studmic, New York, New York *Jun 2020- Dec 2021*

- Responsible for utilizing AI to predict possible collisions before their occurrences, design of electrical control circuitry, and research on sensor fusion with Kalman Filters to perceive the environment.
- Developed algorithms to detect and track vehicles and pedestrians using the OpenCV computer vision library to create state-of-the-art advance collision detection environmental models running on the NVIDIA Jetson Nano.
- Simulated the fusing of a lidar point clouds and camera images; lidar point clouds was clustered and segmented. Utilized RANSAC algorithms for planar model fitting. Using CNN and DeepSORT to improve frame by frame accuracy.

Engineering Assistant, SEAS Electrical Engineering Program, Hempstead, New York *Jan 2020- Feb 2020*

- Built a Light Plate Apparatus in conjunction with research conducted at Cornell aiding in Optogenetics Synthesis.

Electrical Engineering Intern | SBIR, Frequency Electronics, Uniondale, New York *Jan 2019- Dec 2019*

- Co-designed, simulated, programmed, and assembled a preliminary Digital Phase Meter prototype for internal R&D efforts. Design includes a signal conditioning front-end, signal distribution block, and an FPGA for signal processing.
- Developed and tested VHDL signal processing code to measure the beat frequency produced by a test input clock and a DDS signal going into a digital mixer; code transmitted the phase difference of two test input clocks via UART.
- Co-wrote a SBIR proposal for the Department of the NAVY (DoN) on a Multiple Target Phase Synchronization System that utilizes distributed transmit beamforming via UAVs oppose to a single transmitter alternative on a larger Aircraft.

Research Assistant | Autonomy, F.D. School of Engineering, Hempstead, New York *Feb 2018- May 2020*

- Researched drone object detection and its applications to deploy robust footbridges autonomously. Utilized artificial intelligence and video processing to detect objects with a DJI Phantom 4 Pro's video feed.
- Combined computer vision and drones to perform automated tasks in a dynamic environment. Developed control system algorithms for UAV flight using real-time video frame by frame processing.

EDUCATION



Hofstra University, Fred DeMatteis School of Engineering, Hempstead, New York

May 2020

Bachelor of Science, Electrical Engineering

Dean's List, Math Fair Award, Paul & Nancy Summer Science Research Fellowship, Optogenetic Systems Pub.